Notes:

i. All questions are compulsory.
ii. Figures to the right indicate full marks.
iii. Answer to every question must be written on a new page.
iv. L.P.P. problem should be solved on graph paper.
v. Log table will be provided on request.
vi. Write answers of Section – I and Section – II in one answer book.

Section – I

Q.1. Attempt any SIX of the following: **[12]**

i. Express the following statement in symbolic form and write its truth value.
"If 4 is an odd number, then 6 is divisible by 3." (2)

ii. Find the values of x and y, if

$$2\begin{bmatrix}1 & 3\\ 0 & x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$$ (2)

iii. Find the value of 'k' if the function

$$f(x)=\frac{\tan 7x}{2x}, \quad \text{for } x\neq 0$$
$$=k, \quad \text{for } x=0$$

is continuous at $x=0$ (2)

iv. Find $\frac{dy}{dx}$ if $y=\cos^{-1}\left(\sqrt{x}\right)$ (2)

v. The price P for demand D is given as $P = 183 + 120D - 3D^2$.
Find D for which the price is increasing. (2)

vi. Evaluate: $\int\frac{1}{x(3+\log x)}dx$ (2)

vii. If $A=\begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix}$ show that $A^2 - 3A + I = 0$ (2)

viii. Evaluate: $\int x\cos x\,dx$. (2)

Q.2. (A) Attempt any TWO of the following: **[6][14]**

i. Prove that the following statement pattern is equivalent:
$(p \vee q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ (3)

ii. Examine the continuity of the following function:

$$\left.\begin{aligned} f(x) &= x^2 - x + 9, && \text{for } x \leq 3\\ &= 4x + 3, && \text{for } x > 3\end{aligned}\right\} \text{ at } x = 3$$ (3)

iii. Find $\frac{dy}{dx}$ if $y=\tan^{-1}\left(\frac{6x}{1-5x^2}\right)$ (3)

(B) Attempt any TWO of the following: **[8]**

i. Find the inverse of the following matrix by elementary row transformations if it exists.

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 0 & -2 & 1 \\ -1 & 3 & 0 \end{bmatrix}$$ (4)

ii. Find area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (4)

iii. The expenditure E_c of a person with income I is given by $E_c = (0.000035)I^2 + (0.045)\ I$. Find marginal propensity to consume (MPC) and marginal propensity to save (MPS) when I = 5000. Also find A (average) PC and A (average) PS. (4)

Q.3. (A) Attempt any TWO of the following: **[6][14]**

i. Express the truth of each of the following statements by Venn diagram:

a. Some hardworking students are obedient.

b. No circles are polygons.

c. All teachers are scholars and scholars are teachers. (3)

ii. If 'f' is continuous at $x = 0$, then find f(0).

$$f(x) = \frac{15^x - 3^x - 5^x + 1}{x \tan x},\ x \neq 0$$ (3)

iii. Find $\frac{dy}{dx}$ if $x = e^{2t}, y = e^{\sqrt{t}}$ (3)

(B) Attempt any TWO of the following: **[8]**

i. Evaluate: $\int \frac{(1 + \log x)}{x(2 + \log x)(3 + \log x)} dx$ (4)

ii. Evaluate: $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cot x}$ (4)

iii. A firm wants to maximize its profit. The total cost function is C = 370 Q + 550 and revenue is $R = 730Q - 3Q^2$. Find the output for which profit is maximum and also find the profit amount at this output. (4)